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Voltage-temperature health feature extraction to improve prognostics and health management of lithium-ion batteries

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A B S T R A C T

Prognostics and health management (PHM) of lithium-ion batteries are important to ensure the safety of electric vehicles. To date, there has not been an adequate method to accurately estimate battery health conditions and predict battery lifetime under fast charging. A voltage-temperature health feature extraction method is proposed to improve PHM of lithium-ion batteries in this paper. Since voltage change can reflect battery degradation process, a difference model is firstly proposed to extract voltage-dependent health features from partial voltage profiles, which does not need to fully discharge a battery. Simultaneously, as battery aging is affected by temperature, battery surface temperature is selected as a thermal-dependent health feature. Subsequently, the extracted voltage-temperature health features are fed into a developed battery degradation model. Using the proposed method, state of health and remaining useful life (RUL) of lithium-ion batteries can be estimated and predicted with uncertainty measurements. Battery degradation data collected from accelerated battery tests under two different charging policies are utilized to validate the accuracy of the proposed method. Results show that root mean square error (RMSE) is smaller than 1% in all capacity estimation and relative RMSE is around 5% for RUL prediction, which provide higher accuracies than the existing methods.

1. Introduction

Lithium-ion batteries have received much attention during recent years due to their remarkable characters, such as high-energy density, long cycle life and environmental-friendly [1]. With an increase of usage time and charge/discharge cycles, increasingly serious battery degradation may lead to security risks. As a result, battery management system (BMS) is essential to guarantee battery health. Generally, state of health (SoH) and RUL are two primary contents of BMS [2,3]. The accurate estimation and prediction of SoH and RUL have a great significance in avoiding accidents and helping second-hand electric vehicles (EVs) trades in real applications. Much research during recent years has focused on prognostics and health management (PHM) of lithium-ion batteries. However, battery aging is a comprehensive process covering several stress factors, such as high temperature, overcharging/discharging, high pressure, etc. [4], which makes battery diagnosis and prognostics more difficult.

1.1. Literature review

There has been extensive research regarding the SoH estimation and RUL prediction of lithium-ion batteries, which can be broadly divided into three categories: mechanism-based methods, model-based methods, and data-driven based methods [5].

Mechanism-based methods take advantage of the electrochemical knowledge of batteries. This kind of methods enables to model batteries from the inner charge and ion level, according to electrochemical reactions. A classic Pseudo Two Dimensional (P2D) model was established to provide an internal process of charging and discharging to describe battery degradation [6]. Even though

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Nomenclature

Q_{nom}	Nominal capacity of batteries
Q_{cur}	Real capacity in a current cycle
T_{EoL}	The cycle when a battery reaches a pre-defined threshold
T_{ξ}	Current cycle number
V	Discharging voltage
Q_i	Discharge capacity in i th data
f	Functional relationship between V and Q
I	Discharging current
t	Discharging time
$\Delta Q(V)$	The difference capacity
c	Cycle number of a battery
$a(c)$	Offset parameter of c th cycle
$\omega(c)$	Slope feature of c th cycle
$b(c)$	Intercept feature of c th cycle
$\varepsilon(c)$	Normal error of c th cycle
σ^2	Variance of normal error
HF	Health features
$g(\cdot)$	Collection of random variables assumed to follow Gaussian distributions
m_{HF}, $\mathbf{m_{HF}}$	Mean function and mean matrix of $g(\cdot)$
$k_{HF,HF'}$, $\mathbf{K_{HF,HF'}}$	Covariance function and covariance matrix of $g(\cdot)$
$\mathbf{y}'$, $\mathbf{y}$	Predicted and actual health states
$\mathbf{y}'$	Mean prediction results of predicted health states
$\mathbf{K_v}$	Modified Bessel function
$\mathbf{\Theta}$	Hyperparameters of GPR based model

Abbreviations & acronyms

BMS	State of charge
SoH	State of health
SoC	State of charge
RUL	Remaining useful life
EVs	Electric vehicles
SP	Single-particle
P2D	Pseudo Two-Dimensional
KF	Kalman filter
EKF	Extended Kalman filter
UKF	Unscented Kalman filter
PF	Particle filter
HIs	Health indicators
GPR	Gaussian process regression
ANN	Artificial neural network
PHM	Prognostics and health management
ICA	Incremental capacity analysis
DVA	Differential voltage analysis
RVM	Relevance vector machine
V-T	Voltage-temperature
CC	Constant charging
CC-CV	Constant-current and constant voltage
EoL	End of life
DC	Difference capacity
PCC	Pearson correlation coefficient
ME	Maximum error
MAE	Mean absolute error
RMSE	Root mean square error
RMSPE	Root mean square percentage error
SVM	Support vector machine

the accuracy of the P2D model is high, its modeling complexity and computation burden obstruct its potential applications. Afterwards, an equivalent circuit model [7] and a single-particle (SP) model [8] were put forward to simplify the P2D model at the expense of model fidelity. Subsequently, some other improved physical models have been presented. For example, Zhang et al. [9] proposed a modified electrochemical aging model to quantitatively attach internal aging mechanism with battery operation conditions. Han et al. [10] introduced an improved SP model that has higher precision than the original model; also, a simplified P2D model was presented which can estimate the current density distribution and potential of batteries. Unfortunately, mechanism-based methods lack generality for on-board applications because mechanism models are varied from different types of batteries. Notably, the trade-off between computation burden and model accuracy is another critical issue.

Model-based methods utilize available aging data as well as usage times or cycle numbers to model degradation trends by analytical functions [11]. Then, model parameters are updated with fresh operating data by filtering methods such as Kalman filter (KF), extended Kalman filter (EKF), unscented Kalman filter (UKF), particle filter (PF), etc. [12–14]. For example, Shen et al. [15] introduced a two-stage Wiener process model to describe battery degradation in different phases and then utilized unscented particle filter to update model parameters and predict RUL online. Wang et al. [16] conducted a study on battery prognostics under different discharging rates and temperatures to predict RUL with the help of KF and PF. Based on charging cloud data, Li et al. [17] estimated battery life by an Arrhenius empirical model and further improved results by KF and Fuzzy logic. Wang et al. [18] constructed a state-space model of battery capacity and solved it using spherical cubature particle filter. Xu et al. [19] tracked model parameters evolution by using the expectation-maximization and an EKF algorithm and adopted a state-space-based model to predict RUL based on discharging voltage curves. Yang et al. [20] analyzed the relationship between coulombic efficiency and battery aging, and then developed a semi-empirical model that incorporated PF to predict battery RUL under dynamic loading conditions. Although model-based methods are accessible to be implemented for online applications, state-space models comprising observation equations and measurement equations are not easy to design and their estimation accuracy is sensitive to initial parameter values.

Data-driven based methods are attracting widespread interests due to their high precision and model-free characteristics [21,22]. Machine learning approaches have played a key role in battery PHM such as artificial neural network (ANN), support vector machine (SVM), relevance vector machine (RVM), autoencoder [23], etc. Compared to traditional methods, data-driven based methods are convenient to implement. This kind of methods can take advantage of machine learning algorithms and they have no need of physical models or complicated electrochemical knowledge. Besides, the difficulty of getting prior knowledge of state functions in model-based methods can be avoided. In a word, data-driven based methods exhibit the superiority of battery prognostics in the period of big data and have made some new research progress. For example, Shen et al. [24] put forward a deep convolutional neural networks-transfer learning algorithm, taking capacity, voltage and current information as inputs. Liu et al. [25] introduced novel energy-based health indicators (HIs) according to voltage curves of batteries, and then they designed an online SoH estimation framework based on extreme learning machine. Shu et al. [26] developed a least squares-support vector machine approach with a fixed size to conduct SoH estimation using charging times in a predefined voltage range as HIs. Recently, gaussian process regression (GPR) based method is an emerging data-driven approach. Such Bayesian non-parametric probability approach has remarkable performance in nonlinear mapping and it is capable of dealing with issues such as high nonlinearity [27], small sample and prediction uncertainty representation for battery degradation modeling and PHM. Unlike other data-driven methods such as RVM, SVM, ANN, etc., GPR brings statistics thought into machine learning which helps the approach be more accurate. GPR is a non-parametric approach, performing that the input-to-output mapping is regarded as a random function whose probability density is predefined as a Gaussian process in GPR. Its parameter-free character makes it carry out easily and perform better compared to parametric algorithms (ANN, RVM and SVM) [28]. Moreover, GRP can obtain probabilistic prediction and generate uncertainty quantification of prognostics, which is crucial to prognostics. Many works have been published based on GPR. Aiming at forecasting SoH and RUL using limited information, Zhang et al. [29] investigated the relationship between electrochemical impedance spectroscopy and capacity fading and built a GPR based model with automatic relevance determination weights. Electrochemical knowledge was integrated into modified covariance functions, which was creative in this field. Richardson et al. [30] automatically extracted time values between equal-width voltage points as inputs and constructed a GPR model for in situ capacity estimation. Liu et al. [31] took temperature and depth of discharge into account, modified the covariance functions and came up with two innovative GPR-based models.

Unfortunately, although data-driven based methods have made some progress, there remains a need for a more convenient and effective health feature extraction method that can easily extract representative aging features from battery degradation data. General features in the recent literature are extracted based on cycle numbers or voltage curves [32]. Incremental capacity analysis (ICA) and differential voltage analysis (DVA) are two electrochemical techniques that are brought into battery diagnosis and prognostics recently [33,34]. Various features, such as intensities, locations and areas of ICA/DVA curves have been extracted from the curves, and validated to have a high correlation with battery degradation [35,36]. For example, Li et al. [37] extracted peak and valley positions from ICA curve as features of interest and found that plateaus and valleys drift with capacity degradation and can reflect aging mechanisms. Li et al. [38,39] extracted equidistant interpolation points from smoothed incremental capacity curves as features and used GPR-based models to estimate SoH and predict RUL.

However, ICA/DVA have heavy computing burdens and these curves are susceptible to measurement noise. ICA/DVA can only be used under a fully charged/discharged process, which makes applications of ICA/DVA limited. A feature extraction method from partial voltage profiles is needed. Another drawback of such methods is that ICA/DVA cannot reflect the effect of temperature. Furthermore, most existing feature extraction strategies are based on extracting some discrete points in incremental capacity curves or voltage profiles, which are not able to capture the overall characteristics of continuous contours in such curves.

1.2. Contributions of this paper

To solve the aforementioned issues, a novel voltage-temperature (V-T) health feature extraction method is proposed to improve battery SOH estimation and RUL prediction. The main contributions of this paper are summarized as follows. Firstly, since capacity degradation is closely related to voltage change, a difference model aiming to extract health features from partial voltage profiles is constructed. By making difference calculations in voltage profiles, difference capacity curves are derived and subsequently a difference model is constructed by linearizing the curves. Slope feature and intercept feature of the proposed difference model are captured as voltage-dependent health features. Noting that only important information in partial voltage profiles is used so that a fully charged/discharged process is not necessary. Secondly, a novel V-T health feature set is derived. The extracted voltage-dependent features are proved to have a high correlation with capacity according to the Pearson correlation coefficient. Then, by considering a thermal factor which may reflect internal chemical reaction and relate to battery aging, temperature measurement is set as an additional health feature to further improve prediction accuracies. Subsequently, a SoH and RUL forecasting GPR model is proposed to handle an estimate uncertainty problem. The non-parametric GPR-based model is considerably convenient to implement on-board with the support of voltage and temperature measurements. Two series batteries of different charging policies in a commercial lithium-ion battery dataset are used to verify the validity of the proposed method. At last, comparisons with some existing feature extraction methods are conducted to show the better performance of the proposed method over existing methods.

1.3. Organization of the paper

The structure of this paper is outlined as follows. In Section 2, a lithium-ion experimental dataset is introduced. Then, a V-T health feature extraction method is proposed in Section 3. A GPR based method for SoH estimation and RUL prediction is illustrated in Section 4. Experimental results and discussions are made in Section 5. Finally, Section 6 draws some conclusions of this paper.

2. Experimental dataset

In this section, the information of experimental lithium-ion batteries and charging-discharging conditions utilized in this work are introduced.

2.1. Dataset description

The commercial lithium-ion battery degradation data [40] utilized in this work were generated by the Massachusetts Institute of Technology and Stanford University, which aimed to study the effects of fast charging on battery aging. This data set was composed of 124 LFP batteries produced by A123 Systems. The type of these batteries is high power lithium-ion APR18650M1A, and the key technical specification of this kind of batteries is tabulated in Table 1. All the 124 batteries were cycled to 80% of rated capacity under different charging conditions at 30 °C in an environmental chamber (Amerex Instruments). The charging process followed a "R1(S1)-R2 (80%)-1C" policy, where R1 and R2 were two charge C-rates and S1 was the switch state of charge (SoC) between the two constant currents. The aforementioned constant current (CC) stages continued until SoC reached 80%, and finally battery fully charged at 1C CC-CV (constant-current and constant voltage), namely from

Table 1
Technical specification of experimental batteries.

Type	APR18650M1A
Cathode	$LiFePO_4$
Anode	Graphite
Nominal capacity	1.1 Ah
Nominal voltage	3.3 V
Cut-off voltages	3.6 V, 2 V

80% SoC to 100% SOC. It is noted that the C-rate can be calculated by dividing the charge or discharge current by the battery's nominal capacity having the unit of h^{-1}, which is usually used to evaluate the rate of charge or discharge. The cutoff voltages were set at 3.6 V, 2 V, respectively. For instance, "5.3C (54%)-4 C" policy of No. 116 battery means a battery was charged at 5.3 C to 54% SoC and then it was charged at 4 C to 80% SoC. All batteries were discharged at 4 C to 2 V. Raw currents and voltage profiles in a charging and discharging process are given an example in Fig. 1 (a).

In the experiments, surface temperatures of batteries were measured in every cycle. The temperature measurements in the experiments were implemented by attaching a Type T (copper − constantan) thermocouple with thermal epoxy (OMEGATHERM 201) and a Kapton tape to the exposed case of cell by stripping off a small section of the plastic insulation.

Several quantities of batteries including voltage, current, internal resistance, temperature, discharge capacity, etc. Were recorded during charging and discharging process of each cycle. Fig. 1 (b) shows the changes in discharge capacity profiles at different cycles of No. 116 battery in the dataset. The cycle life of No. 116 battery under 5.3 C fast charging policy and 4 C discharging policy is 1157. We just draw the curves every ten cycles here for a clear demonstration. It shows that voltage curves drift with the cycle number increases. In each cycle, firstly, a voltage drops from the upper cutoff potential 3.5 V to around 3.2 V rapidly, and then the rate of voltage decline turns slowly for a long time. Finally, the voltage falls to the lower cutoff potential 2 V much quickly at the end of the discharge process. Clearly, significantly difference occurs among the voltage profiles in different cycles, especially in the later stage, as Fig. 1 (b) exhibits.

Fig. 2 (a) and (b) shows the capacity degradation of two different series of batteries under "4.8 C (80%)-4.8 C" and "5.3 C (54%)-4 C" modes, respectively. The batteries were degraded in different patterns even under the same charging policy and presented various lifetime due to environmental-related factors and cell-to-cell variations.

2.2. Definitions of SoH and RUL

SoH is used to represent battery aging and performance degradation with an increase of usage time. Many quantities can be used as the indicators of battery degradation, such as capacity decline, internal resistance rising and power loss [1]. The SoH is usually defined as the ratio of current capacity over nominal capacity in Eq. (1). Q_{nom} represents the nominal capacity of batteries, which is measured by manufacturers under nominal conditions, including temperature conditions (e.g. room temperature 25 °C), discharging current (e.g. 1C) and so on. Q_{cur} represents real capacity in a current cycle, referring to the maximum amount of charge that battery can be released under a fully discharged process [41].

$$\mathrm{SoH} = \frac{Q_{cur}}{Q_{nom}} \tag{1}$$

Especially, in CC circumstances, current capacity can be obtained by Eq. (2) as follows:

$$Q_{cur} = \int I dt \tag{2}$$

where I is discharging current and t is discharging time.

The above computation equations can only be used in a fully discharged process condition, which is not always satisfied in practice. Besides, the influences of temperatures and other environmental conditions can not be included in Eq. (2) so that the accuracy is difficult to be guaranteed.

RUL indicates the number of cycles from the current state to the end of life (EoL), which is generally set as 80% of an initial capacity under certain charging and discharging protocols [2]. RUL can be defined as:

$$\mathrm{RUL}_{\xi} = T_{\mathrm{EoL}} - T_{\xi} \tag{3}$$

where T_{EoL} is the cycle or time when a battery reaches a predefined threshold, and T_{ξ} is the current observing cycle number. Obviously, SoH focuses on the short-term situation, while RUL concerns more on the long-term status of batteries. Both of them are significant to assess battery health conditions.

3. Extraction of V-T health features based on voltage curves and temperatures

Currently, most existing health feature extraction approaches mainly focus more on point characteristics of IC curves and voltage curves such as capturing time values at equispaced voltages in Ref. [30] or capturing IC values under fixed voltage intervals in Ref. [42]. Besides, some health feature extraction methods are limited to a fully charged state, such as Ref. [43]. To fill the research gap, in this section, we develop a novel V-T health feature extraction approach in virtue of a proposed difference model and measured temperatures. The proposed V-T health feature set consists of two types of components, including two voltage-dependent health features and one thermal-dependent health feature, which will be introduced concretely in the following.

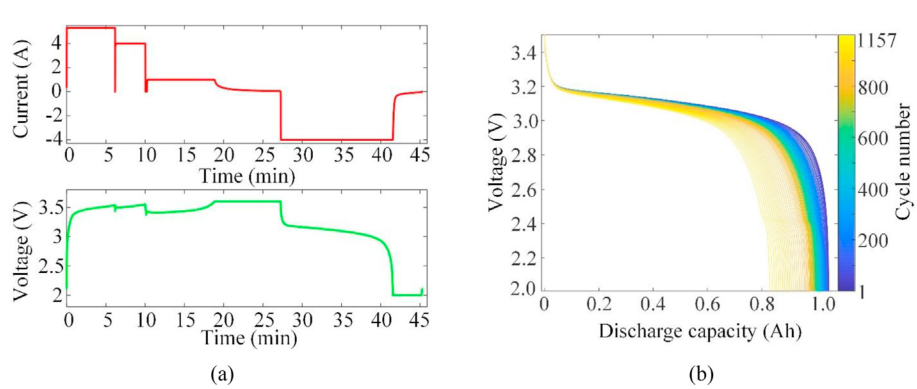


Fig. 1. Dataset representation: (a) A raw current and voltage profile in a charging and discharging process; (b) Voltage profiles v. s. Discharge capacity of No. 116 battery.

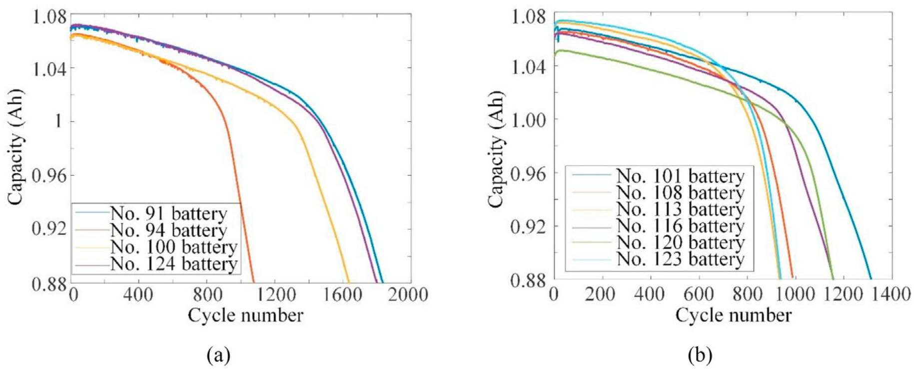


Fig. 2. Battery capacity degradation under two different fast charging conditions: (a) 4.8 C (80%)-4.8 C; (b) 5.3 C (54%)-4 C.

3.1. Voltage-dependent health feature extraction by the proposed difference model

3.1.1. Conventional feature extraction method from IC curves

Characteristics of IC curves can imply internal electrochemical reactions of batteries and reflect an aging degree so that battery health features can be extracted from IC curves [44]. The calculation of IC curves can be expressed as follows. Note that from Eq. (2), discharge capacity is proportional to discharging time because of constant discharging current. The following relationships between discharging voltage and capacity can be acquired:

$$V = f(Q), Q = f^{-1}(V) \tag{4}$$

$$\left(f^{-1}\right)' = \frac{dQ}{dV} = \frac{I \cdot dt}{dV} = I \cdot \frac{dt}{dV} \tag{5}$$

where V and Q are discharging voltage and capacity under the CC stage, and there is a certain functional relationship between them, denoting as f. I is discharging current and t is discharging time.

Generally, Eq. (5) is regarded as an IC curve, which is obtained by the differential calculus. However, differential calculus has several disadvantages such as heavy computation, poor stability, etc. We can imply that if voltage changes slowly, dV is almost zero so that an IC value would be infinite. To address this issue and reduce the computational burden, we extend IC curves by converting the differential calculation to the difference calculation and derive the difference capacity (DC) curves, which can be computed by $\Delta Q(V)_i = Q_{i+1}(V_{i+1}) - Q_i(V_i)$, where i is the serial number of data.

3.1.2. Modified health feature extraction method from DC curves

In this work, we employ the difference calculation to simply the differential operation and to obtain DC curves in Fig. 3. The drop rate of voltage profiles can be demonstrated in this figure by differencing voltage curves, smoothing through the moving averaging algorithm [42] with a parameter of 0.01. The trends that change fast at first and then change becomes slower after reaching a certain range are similar to ecological-evolutionary models, organized by difference equations. Inspired by this, we put forward a difference model to model DC curves.

The strong nonlinear DC curves in Fig. 3 are difficult to analytically model. One feasible method is transferring the curves to linear shapes. Among different cycles, voltage profiles are in a great difference in the later stage rather than the beginning of the discharge process. To model whole voltage profiles in different cycles and exploit valuable health features from voltage profiles, a key point is to transfer the strong nonlinearity of difference voltage profiles to approximate linearity curves. It is noted that only partial voltage profiles will be used to promote the generalization of the method. As we can see in Fig. 3, the curve differences between cycle numbers are mainly located at the certain section around 2.7–3.3 V. However, curves change much steeply on the left side of the curve peaks, which is more difficult to exact related features. Thus, the second half of the curve peak is selected because it can contain the major important information and easy to exact features. After analysis, we set 10^{-3} as the right boundary value of the voltage section. The voltage interval we select is the beginning with peaks of difference capacity and ending less than 10^{-3} value of difference capacity.

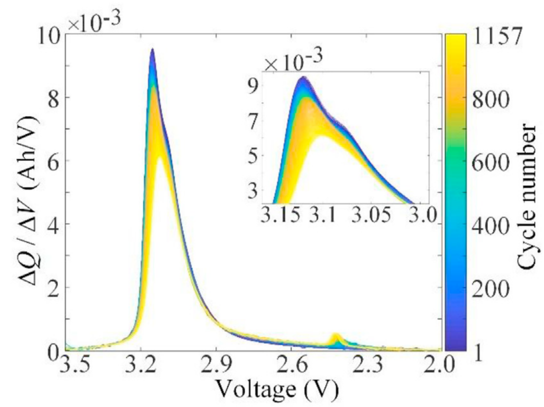


Fig. 3. Difference Q/difference V evolution under different cycles of No. 116 battery.

For setting voltage intervals between two adjacent data points, there is a trade-off between accuracy and calculation amount. The smaller voltage intervals are, the tendencies of voltage profiles are better captured. Smaller voltage intervals will increase computation burdens and computing times. However, If the voltage intervals are too large, then detailed features of voltage profiles may be missed out. After consideration and trade-off, the voltage interval between two adjacent data points we selected is 0.0015 V in this work.

Inspired by the difference equation, we propose a novel difference model to linearize DC curves as follows:

$$\Delta Q(V) = \omega(c)(a(c) - Q(V)) \cdot Q(V) + b(c) + \varepsilon(c) \tag{6}$$

where $\Delta Q(V)$ represents the difference capacity; $a(c)$, $\omega(c)$ and $b(c)$ are respectively model parameters as functions of cycle number c; $\varepsilon(c)$ represents a normal error following $\varepsilon(c) \sim N(0, \sigma^2)$, with 0 mean and σ^2 variance.

Using the proposed difference model, the nonlinear relationship between battery degradation and the underlying information in DC curves can be excavated by the above transformed linear equation. We further derive the slope feature $\omega(c)$ and intercept feature $b(c)$ of the proposed difference model as voltage-dependent health features. Then the characteristics of the entire curves are captured, comparing the existed ICA which can only capture some point features of the curves.

Utilizing such a difference model, the fitting curve and the goodness of fit in one cycle of No. 108 battery are shown in Fig. 4. After fully investigating the fitting performance under different values, $a(c)$ is set to 0 for all cycles of all battery samples. The DC curves are transferred to linear curves with the help of the proposed difference model. Besides, we conduct the hypothesis testing of errors to verify that the assumption of Gaussian error is acceptable. According to the Lilliefors test [45], the p-value of 0.0973 is larger than a significance level of 0.05 to accept the Gaussian distribution assumption of the error term.

Under different cycles, the fitting curves are all linear forms while parameters of fitting curves are various as the charging-discharging cycle number increases. In the next section, we will give the research of the parameter variation trends in detail.

3.1.3. Voltage-dependent health feature extraction

Based on the difference model, the slope features $\omega(c)$ and intercept features $b(c)$ are extracted as new health features, relating to voltage. Fig. 5 (a)–(d) shows the parameter variation trends v. s. The corresponding battery capacity over life cycles. Discharge capacity decreases as the cycle number goes up, degrading to a failure threshold of 0.88 Ah and the two features are the function of cycle number. To better extract features from them, the average moving smoothing method is performed because there are anomaly fluctuations amongst parameters due to the measurement noise of voltages. With cycle number increases, $\omega(c)$ rises monotonically and the rate of rising is similar to the drop rate of capacity. Both of them exist one change-point in the late stage of degradation. Besides, $b(c)$ tends to change in the same direction as capacity. In Fig. 5 (a)–(f), the relationships between $\omega(c)$, $b(c)$ and the capacity of batteries are demonstrated in a three-dimensional space. They can be fitted in a certain three-dimensional surface, which provides the basis for our subsequent modeling.

We can observe that the tendency of discharge capacity and voltage-dependent health features has a high correlation. Pearson correlation coefficient (PCC) is capable of measuring the strength of the linear relationship between two different variables [46]. Using the PCC, we analyze the relationship quantificationally between

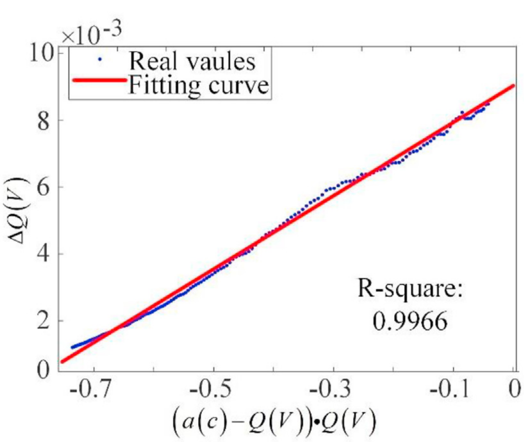


Fig. 4. Fitting results of the proposed difference model.

capacity and the extracted health features. The PCC can be calculated by Eq. (7):

$$r=\frac{\sum_{i=1}^{N}\left(HF_i-\overline{HF}\right)\left(Q_i-\overline{Q}\right)}{\sqrt{\sum_{i=1}^{N}\left(HF_i-\overline{HF}\right)^2}\sqrt{\sum_{i=1}^{N}\left(Q_i-\overline{Q}\right)^2}} \tag{7}$$

where HF are two health features we got above, and Q is discharge capacity in each cycle number i.

The range of PCC is $[-1,1]$, and it is generally acknowledged that the two physical quantities are a perfect positive linear correlation when PCC $=1$, no linear correlation when PCC $=0$ and a complete negative linear correlation when PCC $=-1$. Note that the larger the absolute value of PCC is, the closer the correlation of two physical quantities is to each other.

After calculating, in our study PCCs of $b(c)$ and $\omega(c)$ as well as charging and discharging policies are summarized in Table 2. Although their charging and discharging policies are various, all PCC values of four batteries are close to 1 and -1. Consequently, two voltage-dependent health features have a high correlation to discharge capacity, which means the two features can reflect degradation trends to some extent.

3.2. Thermal-dependent health features

Temperature is a key factor that may affect battery degradation [47]. During charge and discharge process, temperature of battery may fluctuate due to charge/discharge policy switch, the internal chemical reaction and environmental aspect. From an electrochemical viewpoint, temperature changing has some influence on ionic mobility and electrolyte conductivity inside the battery which will influence the aging of battery. Therefore, thermal factor is considered for degradation modeling in our work.

Generally, the thermal factor can be considered by indirect way and direct way. Indirect way is that constructing temperature dependent parameters in a degradation model. However, this way will make model more complicated and be easy to overfit and increase the computational burden in parameter estimation. Also, the effectivity of this way mainly depends on how the temperature dependent parameters are built and their sensitivity to battery degradation, the construction of parameters remains a difficult issue during modeling. Considering the above shortcomings of direct way, we use direct way in this paper, which regards temperature or simple statistics of temperature as health features directly. This way can not only consider the thermal factor in degradation modeling but also keep the model simple.

For comprehensively considering the influence factors of battery estimation and prediction model, we append thermal-dependent measurements to the health feature set as an additional feature. Specifically, in the present work, we select the averaging surface temperatures of a charge/discharge process in each cycle as a thermal-dependent health feature T in Fig. 6. We can see that the general trends of temperatures fluctuate with cycle numbers rise, reflecting the internal reactions.

4. Gaussian process regression related methodology

After extracting V-T health features, we proceed to estimate SoH and predict the RUL of lithium-ion batteries by constructing a battery degradation model.

4.1. Construction of degradation model for batteries

In our work, we construct a battery degradation model to

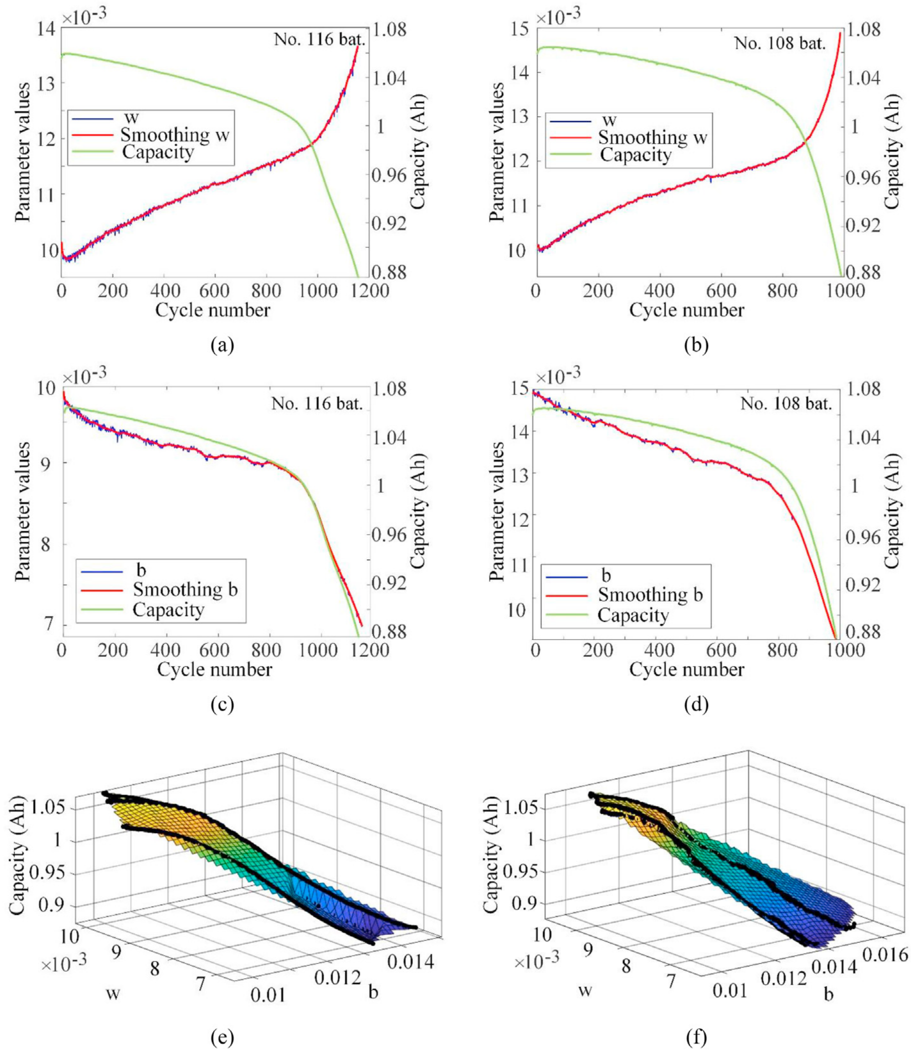


Fig. 5. (a) Variation trends of slope feature $\omega(c)$ and capacity with the cycle number of No. 116 battery; (b) variation trends of slope feature $\omega(c)$ and capacity with the cycle number of No. 108 battery; (c) variation trends of intercept feature $b(c)$ and capacity with the cycle number of No. 116 battery; (d) variation trends of intercept feature $b(c)$ and capacity with the cycle number of No. 108 battery; (e) relationship between $\omega(c)$, $b(c)$ and capacity of batteries under 4.8 C (80%)-4.8 C charging policy; (f) relationship between $\omega(c)$, $b(c)$ and capacity of batteries under 5.3 C (54%)-4 C charging policy.

Table 2
PCCs and charging and discharging policies of four batteries.

Battery number	No. 116	No. 108
Charging policy	5.3 C (54%)-4 C	5.3 C (54%)-4 C
Discharging policy	4 C	4 C
PCC of $b(c)$	0.9945	0.9840
PCC of $\omega(c)$	−0.9461	−0.9409

predict the observation battery health state y_i from V-T health features $\mathrm{HF}_i = \{\omega_i, \mathrm{b}_i, \mathrm{T}_i\}$ and nonlinear mapping function $g(\mathrm{HF}_i)$ in Eq. (8):

$$y_i = g(\mathrm{HF}_i) + \varepsilon_i \tag{8}$$

where the noise term ε_i follows a Gaussian distribution, which can be marked as $\varepsilon_i \sim GP(0, {\sigma_n}^2)$. To solve the degradation model and obtain the battery prognostics, we can make good use of nonlinear mapping approaches such as GPR based model.

We can assume the nonlinear mapping function $g(\mathrm{HF}_i)$ as a collection of random variables which are assumed to follow Gaussian distributions, based on Gaussian process related knowledge. It can be specified by the patterns of various mean and covariance functions as follows:

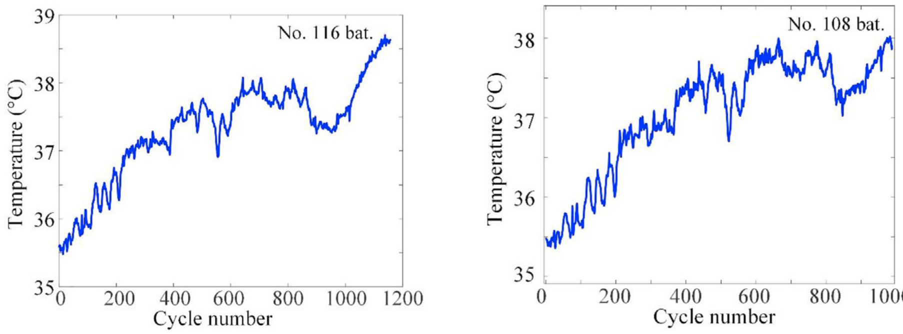
$$g = (g_1, \ldots, g_n)^T \sim GP(m_{\mathrm{HF}}, k_{\mathrm{HF,HF'}}) \tag{9}$$

where $\mathrm{HF}, \mathrm{HF}'$ are two inputs, m_{HF} is a mean function and $k_{\mathrm{HF,HF'}}$ is a covariance function, as shown in Eq. (10):

$$\begin{cases} m_{\mathrm{HF}} = E(g(\mathrm{HF})) \\ k_{\mathrm{HF,HF'}} = E[(m_{\mathrm{HF}} - g(\mathrm{HF}))(m_{\mathrm{HF'}} - g(\mathrm{HF'}))] \end{cases} \tag{10}$$

There are several forms of m_{HF} and $k_{\mathrm{HF,HF'}}$ can be selected. A selection depends on an actual case. The mean function can be set to be zero, and the covariance function, such as squared exponential, is frequently used.

Then for g, a prior distribution follows $p(g(\mathbf{HF})) \sim GP(\mathbf{m_{HF}}, \mathbf{K_{HF,HF'}})$ and by setting $m(x) = 0$, the prior distribution of health state follows:



(a) (b)

Fig. 6. Averaging temperature curves: (a) No. 116 battery; (b) No. 108 battery.

$$\mathbf{y} \sim N\left(0, \mathbf{K_{HF,HF'}} + {\sigma_n}^2\mathbf{I_n}\right) \tag{11}$$

where $\mathbf{HF}$, $\mathbf{HF'}$ are the training set and new dataset that needs to be predicted, respectively; the covariance matrix $\mathbf{K_{HF,HF'}} = (k_{ij})_{n\times n}$ is symmetric and positive definite, calculating through the following matrix (12).

$$\mathbf{K_{HF,HF'}} = \begin{bmatrix} k_{\mathrm{HF_1,HF_1}'} & k_{\mathrm{HF_1,HF_2}'} & \cdots & k_{\mathrm{HF_1,HF}_n'} \\ k_{\mathrm{HF_2,HF_1}'} & k_{\mathrm{HF_2,HF_2}'} & \cdots & k_{\mathrm{HF_2,HF}_n'} \\ \vdots & \vdots & \ddots & \vdots \\ k_{\mathrm{HF}_n,\mathrm{HF_1}'} & k_{\mathrm{HF}_n,\mathrm{HF_2}'} & \cdots & k_{\mathrm{HF}_n,\mathrm{HF}_n'} \end{bmatrix} \tag{12}$$

$$\frac{\partial}{\partial\mathbf{\Theta_i}}\log p(\mathbf{y}|\mathbf{HF},\mathbf{\Theta}) = \frac{1}{2}tr\left(\boldsymbol{\beta}\boldsymbol{\beta}^T - \left(\mathbf{K_{HF,HF}} + {\sigma_n}^2\mathbf{I}_n\right)^{-1}\frac{\partial\left(\mathbf{K_{HF,HF}} + {\sigma_n}^2\mathbf{I}_n\right)}{\partial\mathbf{\Theta_i}}\right) \tag{16}$$

The joint prior distribution of battery health state $\mathbf{y}$ and predicted values $\mathbf{y'}$ is conducted as:

$$\begin{bmatrix}\mathbf{y}\\ \mathbf{y'}\end{bmatrix} \sim N\left(0, \begin{matrix}\mathbf{K_{HF,HF}} + {\sigma_n}^2\mathbf{I_n} & \mathbf{K_{HF,HF'}} \\ \mathbf{K_{HF',HF}} & \mathbf{K_{HF',HF'}}\end{matrix}\right) \tag{13}$$

In this work, the Matérn covariance function [48] is used, which can be expressed as

$$\mathbf{K_{HF,HF'}} = {\sigma_g}^2 \cdot \frac{1}{\Gamma(\nu)\cdot 2^{\nu-1}}\left[\frac{\sqrt{2\nu}}{\rho}|\mathbf{HF}-\mathbf{HF'}|\right]^{\nu}\mathbf{K_v}\left(\frac{\sqrt{2\nu}}{\rho}|\mathbf{HF}-\mathbf{HF'}|\right) \tag{14}$$

where Γ is a gamma function; we set $\nu = 3$, a hyperparameter denoting smoothness of the function; $\mathbf{K_v}$ is the modified Bessel function.

4.2. Hyperparameters estimation and optimization

In Eq. (14), a set of hyperparameters $\mathbf{\Theta} = [{\sigma_g}^2, \rho, {\sigma_n}^2]$ that determine covariance functions should be estimated and optimized. We can optimize hyperparameters by maximizing the log marginal likelihood function as [48]:

$$L(\mathbf{\Theta}) = -\frac{1}{2}\mathbf{y}^T\left[\mathbf{K_{HF,HF}} + {\sigma_n}^2\mathbf{I_n}\right]^{-1}\mathbf{y} - \frac{1}{2}\log\left(\det\left(\mathbf{K_{HF,HF}} + {\sigma_n}^2\mathbf{I_n}\right)\right) - \frac{n}{2}\log(2\pi) \tag{15}$$

$L(\mathbf{\Theta})$ in Eq. (15) can be regarded as a penalized fitness measure, including the data fit term $-\frac{1}{2}\mathbf{y}^T[\mathbf{K_{HF,HF}} + {\sigma_n}^2\mathbf{I_n}]^{-1}\mathbf{y}$, the complexity penalization term $-\frac{1}{2}\log(\det(\mathbf{K_{HF,HF}} + {\sigma_n}^2\mathbf{I_n}))$ and the normalization constant $-\frac{n}{2}\log(2\pi)$.

The gradient-ascent method can be utilized to optimize hyperparameters. The partial derivative of Eq. (15) is expressed as:

with $\boldsymbol{\beta} = (\mathbf{K_{HF,HF}} + {\sigma_n}^2\mathbf{I_n})^{-1}\mathbf{y}$.

Finally, the predicted output $\mathbf{y'}$ can be derived given a new dataset $\mathbf{HF'}$ through computing the posterior conditional distribution $p(\mathbf{y'}|\mathbf{HF'},\mathbf{HF},\mathbf{y})$ in(17):

$$p(\mathbf{y'}|\mathbf{HF'},\mathbf{HF},\mathbf{y}) \sim N(\mathbf{y'}|\mathbf{y'},\mathrm{cov}(\mathbf{y'})) \tag{17}$$

where

$$\begin{cases}\mathbf{y'} = \mathbf{K_{HF,x'}}^T\left[\mathbf{K_{HF,HF}} + {\sigma_n}^2\mathbf{I}_n\right]^{-1}\mathbf{y} \\ \mathrm{cov}(\mathbf{y'}) = \mathbf{K_{HF',HF'}} - \mathbf{K_{HF,HF'}}^T\left[\mathbf{K_{HF,HF}} + {\sigma_n}^2\mathbf{I}_n\right]^{-1}\mathbf{K_{HF,HF'}}\end{cases} \tag{18}$$

In Eq. (18), $\mathbf{y'}$ indicates the mean prediction results of $\mathbf{y'}$, and $\mathrm{cov}(\mathbf{y'})$ is the variance matrix, standing for prediction uncertainty. The 95% confidence evaluation can be computed by $\bar{\mathbf{y}} \pm 1.96\cdot\mathrm{cov}(\mathbf{y'})$.

Constructing the degradation model, the health states such as SoH and RUL of batteries under different charging conditions can be estimated as long as there are available measurements.

4.3. SoH estimation and RUL prediction

The proposed prognostic method comprises two processes, feature extraction process and estimation process. The former part

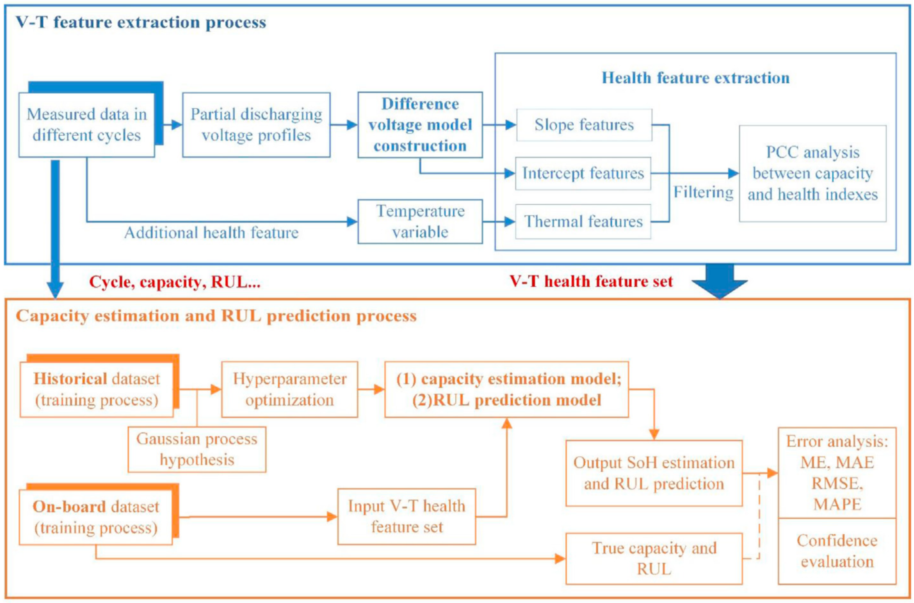


Fig. 7. Flowchart of the proposed method for V-T feature extraction, capacity estimation and RUL prediction.

aims at obtaining a valid health feature set and the latter conducts SoH estimation and RUL prediction using historical data, based on the degradation model introduced before. SoH estimation is implemented by setting capacity as the predicted target while RUL prediction directly sets remaining cycles as the predicted target. Besides, an on-board dataset is employed to verify the effectiveness of the proposed model. Several metrics are utilized to assess the effectivity of the proposed model. Afterwards, estimated results are compared with actual values, then prediction errors and uncertainty measurements are calculated. The flowchart of the proposed method for V-T feature extraction, capacity estimation and RUL prediction is illustrated in Fig. 7.

4.4. Error analysis

The estimation results of the proposed model are assessed by the following general statistical metrics. Maximum error (ME) indicates the max error value of estimation. Mean absolute error (MAE) is a general metric and it emphasizes mean values of absolute errors rather than the symbol of each error. Root mean square error (RMSE) is expected to get larger values than MAE in the same circumstance because it firstly squares the error which magnifies and accumulates the influence of errors. Root mean square percentage error (RMSPE) is similar to RMSE but focuses on the ratio of errors to real values. All metrics are listed as follows.

ME is formulated as:

$$\mathrm{ME} = \max\left(\left|y_i - \widehat{y}_i\right|\right). \tag{19}$$

MAE is formulated as:

$$\mathrm{MAE} = \frac{1}{n}\sum_{i=1}^{n}\left|y_i - \widehat{y}_i\right|. \tag{20}$$

RMSE is formulated as:

$$\mathrm{RMSE} = \sqrt{\frac{1}{n}\sum_{i=1}^{n}\left(y_i - \widehat{y}_i\right)^2}. \tag{21}$$

RMSPE is formulated as:

$$\mathrm{RMSPE} = \sqrt{\frac{1}{n}\sum_{i=1}^{n}\left(\frac{y_i - \widehat{y}_i}{y_i}\right)^2}. \tag{22}$$

where n represents the number of battery samples, y_i and $\widehat{y}_i$ represents the experimental values and the predicted values for sample i.

5. Experimental results and discussions

In this section, we use two series of battery datasets under different charging policies and Fig. 2 shows their capacity degradation. Their charging policies followed 4.8 C (80%)-4.8 C (Case A) and 5.3 C (54%)-4 C (Case B), respectively. Batteries in case A are under constant charging policies, while batteries in case B are charging in a different variable fast charging condition. These two cases can help validate the generalization of our methods and assess battery health status in different fast charging experiments. At the same time, the model sensitivity to the input data is analyzed. We compare the results based on only using historical data and historical & on-board data.

5.1. SoH estimation results

According to the above introduced method, two case studies of SoH estimation are employed. In case A, No. 91 battery and No. 100 battery are training sets and No. 124 battery is applied as a testing set. In case B, using No. 101 battery, No. 108 battery and No. 120 battery as training sets and No. 116 battery is applied as a testing set. The predicted target is to estimate the capacity of the testing

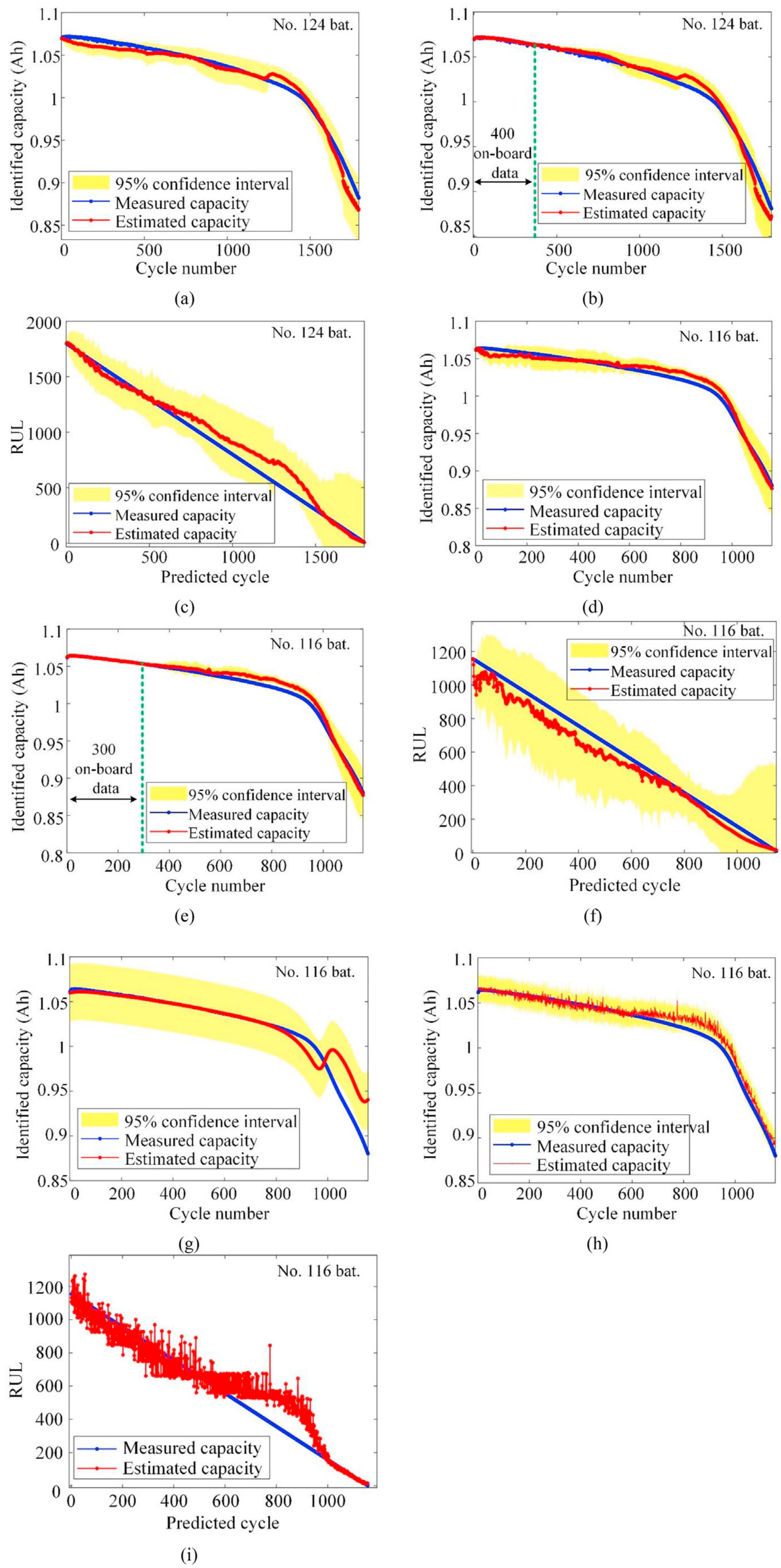

No. 124 bat.
Identified capacity (Ah)
Cycle number
95% confidence interval
Measured capacity
Estimated capacity
(a)
400 on-board data
(b)
RUL
Predicted cycle
(c)
No. 116 bat.
(d)
300 on-board data
(e)
(f)
(g)
(h)
(i)

Table 3
Summary of experimental errors of SoH estimation.

Features	Battery label & data		ME	MAE	RMSE	RMSPE
Proposed V-T health features	Capacity-124 (h)	Historical data	0.0273	0.0056	0.0073	0.76%
	Capacity-124 (h + o)	Historical data, on-board data	0.0259	0.0041	0.0065	0.69%
	Capacity-116 (h)	Historical data	0.0111	0.0056	0.0064	0.63%
	Capacity-116 (h + o)	Historical data, on-board data	0.0112	0.0043	0.0058	0.57%
Cycle numbers	Capacity-116	Historical data	0.0604	0.0078	0.0164	1.76%
IC peaks	Capacity-116	Historical data	0.0278	0.0058	0.0075	0.75%

set. The result is shown in Fig. 8 (a)–(b) and Fig. 8 (d)–(e). For a new battery, there may be limited on-board data we can obtain in practice, so only historical data from the same type battery can be used to estimate capacity. As can be seen in Fig. 8 (a) and (d), the estimated capacity obtained by the proposed method is similar to the measured capacity, with a small RMSPE of estimation is about 0.76% and 0.63%. For on-board SoH estimation, that is considering on-board data of No. 124 battery and No. 116 battery and assuming the battery operates to 400 and 300 cycles respectively, we can get the more accurate estimation results, as Fig. 8 (b) and (e). RMSPE of estimation is declined to 0.69% and 0.57% in this situation. It appears that on-board data can contribute to model improvement in accuracy, and estimation uncertainty is also decreasing with on-board data inputs.

All the experimental errors of SoH estimation are summarized in Table 3. We can see that the proposed model has accurate estimation results in two different cases, and additional onboard data can improve the accuracy of estimation.

To make further comparisons between our proposed method and existing general approaches, the capacity estimation results obtained by cycle number features and IC peak features are diagramed in Fig. 8 (g) and (h). In Fig. 8 (g), cycle numbers are directly used as input health features and capacity estimation is derived by GPR based model. This method achieves a RMSPE of 1.76% compared to 0.63% for our proposed method. Besides, the result shows that the estimation errors tend to be large when approaching the failure of battery, which may cause the problem of accurately judge the battery failure. In Fig. 8 (h), IC peaks are extracted as health features and employed GPR based model to estimate capacity. This method achieves an RMSPE of 0.75%, performing worse than our work. Our proposed method has better performance in capacity estimation against the above two approaches.

5.2. RUL prediction results

In this section, the RUL of lithium-ion batteries is predicted using the V-T health features and GPR related model. About training and testing sets, in case A, No. 91 battery and No. 100 battery are training sets and No. 124 battery is applied as a testing set; in case B, using No. 101 battery, No. 108 battery and No. 120 battery as training sets and No. 116 battery is applied as a testing set. The inputs of RUL prediction are V-T health features too, similar to SoH estimation. However, the predicted target is remaining cycles respect to cycle life of batteries in RUL prediction, which is different from SoH estimation. RUL in each cycle will be predicted in this process and prediction uncertainty will be performed as the form of 95% confidence interval.

As illustrated in Fig. 8 (e) and (f), the RMSEs of RUL prediction results of No. 124 battery and No. 116 battery are 93.27 and 64.06 respectively, which is 5.17% and 5.54% of their actual lifetimes. The performance of the proposed model is notably acceptable in the condition of limited inputs.

Comparative experiments are conducted in RUL prediction, shown in Fig. 8 (g)–(h) and Table 3. To compare the effectivity of our recent studies to related works, we select other previous methods such as using IC peaks as health features to predict RUL of batteries. This approach achieves a RMSE of 103.07 compared to 64.06 for our method, shown in Fig. 8 (i). And the error is 8.92% of their actual lifetimes. It is clear that our method outperforms previous methods, therefore the effectivity of the proposed method in RUL prediction can be proved.

6. Conclusions

BMS is an important part in the EVs business. Plenty of prior works have paid great attention to PHM of lithium-ion batteries to guarantee the security of EVs. However, these studies rarely focus on health feature extraction. In this work, a voltage-temperature health feature extraction method was proposed to extract effective features to improve prognostic and health management of lithium-ion batteries. Two voltage-dependent health features were constructed from partial voltage profiles, with the help of the proposed difference method. Meanwhile, a thermal-dependent health feature was extracted to consider the effect of temperature on battery aging. The extracted voltage-temperature health features were then fed into a GPR based model to estimate the SoH and RUL of lithium-ion batteries. The data collected from two series of batteries datasets under different fast charging conditions were used to validate the accuracy and robustness of the proposed method. Results showed that the proposed method is effective and robust to capacity estimation, with all RMSEs smaller than 1%. Moreover, the proposed method can provide effective RUL prediction, with RMSE around 5%. Through the comparisons, our proposed method was validated to be more accurate than some general existing feature extraction methods.

To summarize, this work provides a convenient voltage-temperature health feature extraction method for battery performance evaluation method under fast charging testing, which can help optimize a fast-charging strategy in order to get a balance between short charging time and good battery performance. In our future work, more types of batteries will be analyzed to demonstrate the generalization of our proposed method.

Fig. 8. Experimental results: (a) capacity estimation of No. 124 battery on historical data using the proposed method; (b) capacity estimation of No. 124 battery on historical data and on-board data using the proposed method; (c) RUL prediction of No. 124 battery using the proposed method; (d) capacity estimation of No. 116 battery on historical data using the proposed method; (e) capacity estimation No. 116 battery on historical data and on-board data using the proposed method; (f) RUL prediction No. 116 battery using the proposed method; (g) capacity estimation of No. 116 battery using general cycle features; (h) capacity estimation of No. 116 battery using IC peaks; (i) RUL prediction of No. 116 battery using IC peaks.

Author contribution

Jin-zhen Kong: Conceptualization, Methodology, Formal analysis, Investigation, Project administration, Validation, Visualization, Writing − original draft, Writing − review & editing. Fangfang Yang: Writing − review & editing, Validation. Xi Zhang: Resources, Investigation. Ershun Pan: Project administration, Funding acquisition. Zhike Peng: Project administration, Funding acquisition. Dong Wang: Writing − review & editing, Supervision, Funding acquisition.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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